

No evidence for Nonconscious Goal Pursuit in Novel Environments: A Replication of Eitam
et al.'s (2008) Experiment 2

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Abstract

In two experiments, Eitam, Hassin, and Schul (2008) found evidence that priming of nonconscious goal pursuit could extend beyond reproductions of past associations and facilitate novel learning, specifically implicit learning. Given the implications of such findings and dearth of subsequent investigations of this phenomenon, we conducted a direct replication of their second experiment using materials graciously shared by the original authors. Despite using a sample size that provided .80 power to detect the original effect size, we failed to find evidence that priming of nonconscious goal pursuit facilitated implicit learning. Our replication was conducted as part of the Reproducibility Project: Psychology (Open Science Collaboration, 2015) and study preregistration, materials, and data are available online (<https://osf.io/edcr7>).

Keywords

Implicit learning; nonconscious goal pursuit; priming; replication; reproducibility

1. Background

Eitam, Hassin, and Schul (2008, p. 261) asked the question: ‘To what degree can people count on unconscious processes to pilot them toward the future they desire, that is, toward their goals?’ Dual-system models have proposed that there are two distinct cognitive systems involved in human reasoning: an associative one that is automatic and more domain-specific and one that is more deliberate and might involve “constructing mental models or simulations of future possibilities” a process that has been called hypothetical thinking (Evans, 2003, p. 454).

The unconscious processes, or nonconscious goal pursuit (NCGP), that Eitam and colleagues are referring to in their question above had typically been assumed to only play a role in the associative reasoning system. Thus, NCGP would only be helpful in facilitating goals through past associations that were not intentional or conscious (e.g., Kruglanski et al., 2002). Eitam and colleagues highlighted that consistent with this, the existing research on NCGP focused on domain-specific goals that had strong existing associations. One example of this is the finding that participants are more likely to choose a low-fat food option when the goal of dieting was nonconsciously primed (Fishbach et al., 2003).

Eitam, Hassin, and Schul (2008) noted the limited usefulness of NCGP if it could only impact existing, overlearned associations and wanted to see if NCGP could facilitate novel learning, specifically implicit learning. Since implicit learning involves learning something without intending to and without awareness, such learning could not be the result of activating past learning associations. Eitam and colleagues performed two separate experiments where participants were randomly assigned to either a goal-priming (exposed to goal-directed words) or control condition (exposed to neutral words not related to goal achievement). In both experiments they found that those participants in the goal-priming conditions outperformed those in the control conditions on implicit learning tasks even

though the NCGP priming words were not necessary or needed for accurate performance on the implicit learning task. Thus, they demonstrated that it was possible for NCGP to facilitate learning in novel environments.

The only attempt to replicate the findings of Eitam and colleagues (2008) that we have been able to locate was Gamble et al.'s (2014) study which used a comparable goal priming task, but looked at learning on an Alternating Serial Reaction Time (ASRT) task as opposed to the Serial Reaction Time (SRT) task used by Eitam and colleagues. Gamble and colleagues indicated that they used the ASRT in order to address two limitations of the original study. First, they indicated that the ASRT might be a better measure of implicit learning than the SRT as participants are unable to differentiate the sequence they've unconsciously learned from those that they haven't even after they have received extensive training on the sequence (Howard et al. 2004). They argued that even though Eitam and colleagues did not see a difference in explicit knowledge between their goal-priming and control condition, they did not examine whether or not the explicit knowledge displayed was greater than that expected by chance. Thus, it is possible that it was explicit, and not implicit learning that was being assessed by Eitam and colleagues with the SRT. Second, unlike the SRT which only assessed learning at one time-point (after the four training blocks were administered), the ASRT assesses learning during training, which enabled Howard and colleagues to compare learning in the goal-priming and control conditions throughout training.

In this replication of Eitam et al.'s (2008) experiment 2, the ASRT repeated the pattern to be learned ten times per block and participants experienced fifteen blocks (see Gamble et al., 2014 for additional details). To examine the time-course of learning, and any differences in learning between the goal-priming and control conditions, Gamble et al. divided the fifteen blocks into 5 epochs of time, with each epoch being made up of 3 blocks

(no justification was provided for creating 5 epochs in this manner). Because Gamble et al. looked at different epochs, their examination of learning differences between the two groups involved testing the main effect of group (goal-priming versus control) as well as a 3-way interaction of group by triplet type by epoch; the response time comparison of high to low probability triplet types was used as the indicator of sequence learning (see Gamble et al. for details). Although neither the group ($p > .10$) nor group by triplet type by epoch ($p = .07$) achieved statistical significance, the authors used the .07 as evidence of a 'marginal' interaction that justified comparing the group by triplet type interaction at different epochs. Although none of the interactions were significant for any of the 5 epochs, the authors indicated that they found a "trend toward learning differences between the two groups" (Gamble et al., p. 3639) such that the goal-priming condition showed faster responding (i.e., enhanced implicit learning) in the second epoch ($p = .067$), whereas the control condition showed faster responding in the fourth epoch ($p = .077$).

Gamble et al. (2014) conducted a second experiment which was the same as the first except they had 15 epochs (45 blocks) of the ASRT task instead of 5 epochs. Gamble and colleagues indicated that this was to "replicate these findings, and also to allow extended training in the ASRT to better determine the time course of the goal motivation effect" (p. 3640). This time they did find a significant group by triplet type by epoch interaction, $p = .03$, with post-hoc tests revealing enhanced implicit learning in the goal-priming condition during the fourth epoch ($p = .006$), enhanced implicit learning in the control condition during the sixth epoch ($p = .02$), and no significant differences in implicit learning between the goal-priming and control groups for any other epoch.

Given that there has never been a study that attempted to replicate the exact methods of Eitam, Hassin, and Schul's (2008) second experiment, and that there were somewhat mixed findings for the one study that did a replication with a different outcome measure, the

present study sought to do as close to an exact replication as possible to see if the finding that NCGP enhances implicit learning could be replicated. Experiment 2 was the one replicated because this study was conducted as part of the Reproducibility Project: Psychology, which asked researchers to replicate the final study in an article (if there were multiple studies). Although a sample size was collected to yield a power of .80 to detect the original effect size when running the same analyses as Eitam and colleagues, additional analyses were conducted which resulted in 99.5% power to detect the original effect. In an effort to match the procedures and materials of the original study as closely as possible, we consulted with the original authors and asked for their materials; all materials requested were graciously provided.

1.1. Social Impact & Responsibility

If NCGP can enhance implicit learning, this has potential implications for society with regards to how people might enhance their ability to work towards goals. However, the use of nonconscious goal priming also carries potential risks; for example, people engaging in phishing attacks might use NCGP to activate somebody's goals related to financial outcomes which might make them more susceptible to being scammed by the phishing attack. However, many of the potential risks are related to NCGP more generally and not specific to the research hypothesis being replicated here, which is specific to NCGP and implicit learning.

1.2. Original and Replication Study References

	APA-formatted reference with DOI
Original	Eitam, B., Hassin, R. R., & Schul, Y. (2008). Nonconscious goal pursuit in novel environments: The case of implicit

	learning. <i>Psychological Science</i> , 19, 261-267. https://doi.org/10.1111/j.1467-9280.2008.02078.x
Replication	Prenoveau, J. M., & Kirkhart, M. W. No evidence for Nonconscious Goal Pursuit in Novel Environments: A Replication of Eitam et al.'s (2008) Experiment 2. https://osf.io/edcr7

1.3. Replicated Study and Hypothesis

While the original article conducted and reported two experiments, we replicated Experiment 2, 'Unintentional Learning'. The main hypothesis of this experiment that we were re-testing was that participants who were primed for NCGP would show greater learning on an implicit learning task (the serial reaction time, or SRT, task) than those in a control condition who were not primed for NCGP.

2. Methods

2.1. Proposed Materials and Procedures

This section details the proposed methods and procedures that were originally specified in the preregistration of this replication. Preregistration and materials can be found at <https://osf.io/edcr7>. Deviations from these materials and procedures are documented in Section 2.2 below.

Materials

Goal priming task. The NCGP priming task was a word-search puzzle. An 11 x 11 array of letters comprised the word-search task, within which the participants were asked to locate 13 target words. Six words were the same for both conditions (*blue, building, candle, chair, stamp, and tree*). For the goal-directed condition seven of the words were associated

with achievement (*ambitious, competition, excellence, first, goal, racing, and win*). For the control condition seven motivationally neutral words of approximately the same length were used instead (*carpeting, diamonds, farm, hat, limestone, table, and window*).

Priming awareness probes. Awareness probes were used to assess the extent to which “participants believed the cover story, and whether they had any suspicion regarding the priming manipulation. First, they were asked to speculate, in their own words, on the ‘purpose of the word-search task’ (which served as the priming manipulation). Next, they were asked ‘whether there was anything noteworthy in the word-search task,’ and last, they were asked directly whether the word-search task had affected their performance” (Eitam et al., 2008, p. 263) on the SRT task.

SRT task. A widely used incidental-learning task, the serial reaction time task (SRT; Nissen & Bullemer, 1987) was used as a measure of implicit learning. It is widely accepted that participants can demonstrate significant learning of complex sequences in this task without intention or awareness of the predictability of the sequences.

DirectRT software was used to deliver the SRT task. An SRT block was made up of 50 trials; an SRT trial consisted of a circle being displayed to participants on a computer screen in one of four possible locations along a horizontal line (Eitam et al., 2008).

Participants were told to record the location of the circle by pressing the number key which corresponded to the circle’s relative location on the screen: 3 for the far left location, 5 for the second to the left location, 7 for the second to the right location, and 9 for the far right location. As soon as participants pressed one of these keys, the circle vanished from the screen and 200 ms later, another circle appeared at one of the four locations. If the participant responded incorrectly, a red ‘X’ would appear in the middle of the screen; the X and displayed circle would both continue to be displayed until the correct number key was entered (Eitam et al., 2008, p. 264).

During the first block (50 trials), the circle location was random (Eitam et al., 2008, p. 264). This practice block allowed participants to get used to doing the SRT task and the circle locations were randomized because if they wouldn't think there was a specified sequence, they would be less likely to attempt to try to explicitly learn a sequence. After the practice block, participants received four training blocks: the order of circle locations was fixed at 3759757975 and this sequence was repeated five times in each block (Eitam et al., 2008).

After these training blocks, participants were administered three 'test' blocks (Eitam et al., 2008). The first and third of these blocks, which Eitam et al. called 'random' blocks, involved random presentation of circle locations with the following two constraints: the circle appeared at the same frequency in each location in the random blocks as in the fixed-sequence training blocks and the circle never consecutively appeared in the same location. Circle locations in the second test block were presented in the same order as that of the four earlier training blocks (Eitam et al., 2008). As described below in the Analysis Plan section, reaction time is used as an indicator of incidental learning by comparing reaction times during the test fixed block to reaction times during the test random blocks.

SRT task awareness probes. To determine if participants were aware that there was a repeating fixed sequence of circle locations during certain blocks, they were asked to describe the circle location sequence in their own words and these descriptions were scored using a modified version of Seger's (1997) scoring (Eitam et al., 2008). Participants were then informed that the circle locations had been displayed in a fixed order and they were asked to indicate the circle location order for a sequence of 10 locations. Participants were assigned a knowledge score ranging from 0 (no knowledge) to 10 (full knowledge) based on a scoring algorithm that compared their sequence to that of the sequence used in the training blocks (Eitam et al., 2008).

Self-reported explicit achievement motivation. Participants completed three Likert-type items on a 9-point scale ranging from 1 (*not at all important*) to 9 (*very important*) to assess their explicit achievement motivation: how important it was for them to respond quickly, how important it was for them to respond accurately, and how important it was for them to succeed. (Eitam et al., 2008, p. 264).

Self-reported intention to learn sequence. Using the same Likert-type scale described above, participants were asked how important it was for them to learn the sequence of the circle locations (Eitam et al., 2008).

Procedure

All participants were provided with a consent form prior to participation. A research assistant reviewed the key points of the consent form with participants and answered any questions they had as part of the informed consent process. The consent form detailed the fact that participants would be completing two unrelated experiments, with the first – a word-search task - being described as a pilot experiment for a task that will be used in future research. The second experiment, the ‘main study’ was described as a cognitive investigation of college student reaction times.

Block randomization was used to randomly assign participants to either the goal-priming or control condition (Eitam et al., 2008). Participants in both conditions were given a paper handout with an 11 by 11 letter array and were instructed to find and circle 13 words. As detailed above, seven of the words in the goal-priming condition were associated with achievement; in the control condition, these were replaced with seven motivationally neutral words of approximately the same length (Eitam et al., 2008).

Participants were thanked after completing the word-search task and were then brought to a computer station and asked to complete the ‘second experiment’ (Eitam et al., 2008). Participants then completed the eight blocks (fifty trials each) of the SRT task

described above: a practice block (random circle location), four training blocks (fixed circle location), a random test block (random circle location), a test block (fixed circle location), and a random test block (random circle location). Participants were instructed to press the number key corresponding to the circle location; no information was provided about the learnable fixed pattern (Eitam et al., 2008).

Upon completion of the eight SRT task blocks, participants were administered the SRT task awareness probes, self-reported achievement motivation items, self-reported intention to learn sequence item, and priming awareness probes (Eitam et al., 2008).

Analysis Plan

Participants' data were excluded from the analyses if they demonstrated awareness of the motivation link between the goal-priming word-search task and the SRT task. In addition, participants' whose learning performance on the SRT task was 4 standard deviations from the mean were also excluded.

SRT learning. The index of SRT learning was calculated by subtracting the average response time of the two random blocks from the response time of the fixed block. Thus, larger negative difference scores are indicative of greater learning. Typically, only correct responses are used to assess learning differences; although it is not explicitly stated in the original manuscript, it was confirmed with the authors that this is what was done in the original study, so that is what was done in the replication. The mean difference score for the control condition was then compared to the mean difference score for the experimental condition using an independent samples *t*-test. Likewise, to assure that the differences observed between the control and experimental condition was due to improvements in learning the fixed sequence, the means for the two random blocks of trials were compared for the two conditions using an independent samples *t*-test. Finally, as done in the original study, mean reaction times for the experimental and control conditions were compared regardless of

type of test block. An index of SRT learning using the median, rather than mean, response times was also calculated because the original article indicated that an analysis using the median response times produced an identical pattern of results to that of the mean response times.

Explicit achievement motivation. A series of *t*-tests were conducted to determine if the control and experimental groups differed on their explicit motivation when performing the SRT task. Specifically, the mean answers on the post-experimental questionnaire about performing the SRT task accurately, quickly, and general desire to succeed at the task were each compared with a *t*-test. Intention to learn the fixed sequence was also compared between the two groups with a *t*-test.

Explicit knowledge of the SRT task. Additional *t*-test analyses were conducted for the two measures of explicit knowledge about the SRT task, specifically the degree of accuracy of describing the SRT task in their own words, and the accuracy of their graphic predictions of the fixed sequence in the SRT task.

Additional Analyses. Although not done in the original study, the authors' preregistration proposed doing additional analyses for the replication in an a priori manner. Specifically, the design of this study lends itself well to using a repeated-measures ANOVA to test the effect of the manipulation on implicit learning (as compared to the multiple *t*-tests used in the original manuscript). Therefore, the authors proposed to examine the impact of the manipulation on implicit learning by using a 2 (condition: experimental vs. control) x 2 (test block type: fixed vs. random) ANOVA looking at response times, where condition is a between-subjects factor and test block type is a within-subjects factor. The question under investigation would be examined by testing the interaction of condition and block type: one would expect a significant interaction such that those in the experimental condition show a significantly larger decrease in response time when comparing random to fixed blocks than

those in the control condition. If the interactive effect is as large as the observed effect size in the original manuscript (*Cohen's* $d = .451$ translates to an effect $f = .226$), then for an alpha of 0.05, a correlation among measures of 0 (most conservative possible), and a nonsphericity correction of 1, the planned sample size of 158 would yield a power of 99.5%.

A Priori Sample Size Determination

The effect size for the finding that has been targeted for replication is a Cohen's d of .451. Consistent with the original study, a two-tailed test with an alpha of .05 was used. Assuming an equal number of participants in each group, a sample size of 158 participants are needed to achieve a power of 80% to detect an effect this large or larger. Although this sample size was targeted to give a power of 80% to detect an effect size as large, or larger, than that found in the original study, additional analyses were proposed which resulted in 99.5% power to detect this effect with a sample size of 158 (see additional analyses above).

Sample

As with the original study, participants were undergraduate university students recruited from an Introduction to Psychology course pool of volunteers who received course credit for participation, or were from the general student body who received nominal monetary compensation for participation. Students were recruited from Loyola University Maryland, a private, liberal-arts school that is located in the Mid-Atlantic region of the United States. at the time of recruitment, there were approximately 4,000 undergraduates, roughly 60% were female, and the ethnic distribution of the students at Loyola as determined by self-report was approximately: 78% Caucasian, 10% Hispanic, 4% African American, 3% Asian, 3% two or more ethnicities, less than 1% American Indian / Alaskan Native, and less than 1% Pacific Islander / Hawaiian Native. The majority of students, roughly 73%, came from the Middle States region, which includes Delaware, the District of Columbia, Maryland, New Jersey, New York, and Pennsylvania.

Given the target sample size of 158, 79 participants with usable data were needed in each condition. Because the data was not scored and analyzed during the data collection time period, and because those scoring the data were blinded to condition, data was collected for 206 participants with the expectation that this would provide at least 79 usable cases in each condition and the first 79 usable cases in each condition would be used for the analyses in order to provide a power of .80 based on the original effect size.

Of the first 79 participants run in the goal-priming condition, 5 had to be excluded. One participant had to be excluded because it was suspected that they were aware of the motivation link between the goal-priming word-search task and the SRT task. The other four participants had to be excluded for technical reasons (see section 2.2 Deviations from the Proposed Materials and Procedures below for details). Thus, goal-priming participants 80 through 84 were included in the sample. Of the first 79 participants run in the control condition, 4 had to be excluded for technical reasons. Thus, control participants 80 through 83 were included in the sample. Two participants, one in each condition, had SRT learning scores (-486.5 and -472.9) that were 4 or more standard deviations from the mean of the SRT learning scores of the 158 participants ($M = -33.4$, $SD = 72.5$); as done in the original paper, negative numbers indicate learning: faster times in the fixed sequence block relative to the two random blocks. The next participant in the goal-priming condition had technical difficulties so the subsequent participant was included to bring the sample size of the goal-priming condition to 79. The next participant in the control condition was included to bring the control condition sample size to 79. Thus, data was used for 79 of the first 86 participants in the goal-priming condition; 5 were excluded for technical reasons, 1 was excluded because it was suspected that they were aware of the motivation link between the goal-priming word-search task and the SRT task, and 1 was excluded because their SRT learning score was 4 standard deviations from the mean of the SRT learning scores. Data was

used for 79 of the first 84 participants in the control condition; 4 were excluded for technical reasons and 1 was excluded because their SRT learning score was 4 standard deviations from the mean of the SRT learning scores.

2.2. Deviations from the Proposed Materials and Procedures

Although the originally proposed and preregistered method plan did not explicitly state that participant age and gender would be assessed, it could be inferred that they would be assessed by the fact that age and gender of the sample was discussed. However, we forgot to assess this information and did not realize they we were not collecting it (when we should have been) until after the first 115 participants had been run. At this point, questions to assess gender and age were added to the very end of the procedure (so that subsequent participants' experience of the remainder of the experiment would be no different from the first 115 participants). After IRB approval was obtained, gender and age data was sought from the first 115 participants by sending them a follow-up email. However, not all participants responded to provide this information. For the 115 of 158 who did provide gender and age data, 74.8% ($n = 86$) were female and the mean age was 19.65 ($SD = 1.1$).

Although 'technical reasons' were not given as exclusionary criteria in the original methods, it was deemed necessary to include technical reasons as an exclusionary criterion after we reviewed all participant data. Of all those excluded for technical reasons, the majority of participants excluded did not have response time data for the SRT task because the task shut down before the relevant blocks were recorded. Therefore, they could not be included in the key analysis (SRT learning) because they did not have data for this variable. One participant did have data for these blocks, but was excluded because they had already run part-way through the SRT task before it shut down. After the task shut down, the research assistant restarted the task at the beginning. Therefore, this participant's learning

history would be different than that of all the other participants (they underwent more training blocks than other participants).

A couple of the variables assessed required subjective assessment of the responses provided by participants (e.g., knowledge of the motivation link between the word-search task and the SRT task, degree of accuracy of describing the SRT sequence in their own words). The original method section did not provide detail as to how these variables would be assessed. The scoring protocol for these (and other) variables are provided at <https://osf.io/edcr7>. These variables were first scored by research assistants who were blind to condition during scoring. The variables assessing suspected knowledge of the motivation link between the word-search task and the SRT task were then verified by the two primary researchers who were also blind to condition while scoring participant responses.

2.3. Original and Replication Study Features

Table 1

Differences Between the Original and the Replication Study.

Study Feature	Original Study	Replication Study	Justification and Possible Effect on Interpretability
Data collection time	Prior to 2009	10/2013 – 04/2014	Does not apply.
Language of written materials (e.g., consent form, word-search task)	Hebrew	English	Participants in replication were English.

Sample	Hebrew University undergraduates (Israel)	Loyola University Maryland undergraduates (United States)	Participants in replication were recruited from authors institution.
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2.4. Replication Closeness

Table 2

Replication Closeness According to Features Proposed by Brandt et al. (2014)

Study Feature	Closeness rating: exact, close, different, or does not apply.	Justification for closeness rating
Instructions	Close	Replication materials were translated into English from the original Hebrew materials.
Measures	Close	Replication materials were translated into English from the original Hebrew materials.
Stimuli: Word-search	Close	Replication materials were translated into English from the original Hebrew materials.
Stimuli: SRT task	Exact	The program and image(s) displayed as part of the task was the same in the replication as in the original.

Procedure	Exact	The procedures described in the original study, as well as the additional details provided by the original authors, were precisely and exactly followed in the replication.
Location (e.g., lab vs. online, alone vs. in groups)	Exact	Both original and replication were conducted in person in a laboratory setting and participants were run individually. Replication lab set-up was based on detailed descriptions from original authors.
Remuneration	Exact	Participants in both original and replication could obtain pay or course credit.
Participant populations	Close	The target population in both original and replication were university undergraduates. However, the original sample was 60% female with an average age of 23.5 years whereas the replication sample was 75% female and the mean age was 19.65.

Participant exclusions / exclusion criteria	Close	All exclusion criteria from the original study were followed in the replication study. However, as noted above, the replication study also excluded participants for technical reasons (e.g., did not have complete data for the dependent variable) whereas this was not noted as an exclusion criteria in the original study.
Language	Different	Hebrew (original) and English (replication).
Nationality of the participants	Different	Recruited from a university in Israel (original) and the United States (replication).

2.5. Involvement of Original Authors

Drs. Hassin and Eitam were contacted during the design stage of the replication and they were incredibly generous in sharing materials and helpful in answering questions. As far as we can tell, they provided us with all the materials that they used for the original study, including the coding used for the SRT task, the written materials provided to participants, and the instructions guiding the researchers who were carrying out the protocols. They also answered all questions that we put to them about how the experiment was carried out, the laboratory set-up, etc.

2.6. Sample Size Justification

As detailed above, the target sample size was 158 participants (79 per condition) to achieve 80% power to detect the original effect size with a two-tailed test for an alpha of 0.05. As also noted above, this sample size provided a power of 99.5% when the hypothesis of interest was tested by examining the interaction of a repeated-measures ANOVA.

2.7. Preregistration

	Link for preregistration	Type of template
Original		
Replication	https://osf.io/edcr7	Replication Recipe

3. Results

SRT learning

Figure 1 presents the mean RTs of correct responses in the three test blocks. The average learning score (average fixed block response time minus the average response time of the two random blocks) for those in the goal priming condition ($M = -22.3$, $SD = 51.3$) was not significantly different from those in the control condition ($M = -33.1$, $SD = 52.1$), $t(156) = 1.318$, $p = .189$, $d = -0.21$. When averaging across all participants and all three blocks, the average accuracy was 93.3%: out of 150 trials, the mean number of correct trials was 139.96 ($SD = 8.65$). The average accuracy did not differ between groups for any of the three blocks, all $ts(156) < 1.71$.

Although the average SRT learning score for those in the goal priming condition was not significantly different from those in the control condition, response time performance in the two random blocks was still compared, as done in the original study, to determine if one condition showed enhanced performance in general (not necessarily related to learning). No

significant differences were found between the goal priming and control conditions when using either mean or median response times, all $t(156) < 1.25$.

Explicit achievement motivation

The average score for explicit motivation to perform the SRT task accurately did not differ significantly between the goal priming condition ($M = 7.6$, $SD = 1.3$) and the control condition ($M = 7.7$, $SD = 1.1$), $t(156) = -.514$, $p = .608$, $d = 0.08$. The average score for explicit motivation to perform the SRT task quickly did not differ significantly between the goal priming condition ($M = 7.2$, $SD = 1.4$) and the control condition ($M = 7.2$, $SD = 1.4$), $t(156) = -.229$, $p = .819$, $d = 0.04$. The average score for explicit motivation to generally succeed at the task did not differ significantly between the goal priming condition ($M = 7.3$, $SD = 1.5$) and the control condition ($M = 7.4$, $SD = 1.3$), $t(156) = -.409$, $p = .683$, $d = 0.07$. The average score for explicit motivation to learn the fixed sequence did not differ significantly between the goal priming condition ($M = 5.5$, $SD = 2.5$) and the control condition ($M = 5.0$, $SD = 2.7$), $t(156) = 1.354$, $p = .178$, $d = -0.22$.

Explicit knowledge of the SRT task

Explicit knowledge of the fixed sequence order as measured by the degree of accuracy in describing the SRT task in their own words did not differ significantly between the goal priming condition ($M = 1.9$, $SD = 1.2$) and the control condition ($M = 2.1$, $SD = 1.2$), $t(156) = -0.668$, $p = .505$, $d = 0.11$. Explicit knowledge of the fixed sequence order as measured by their graphic predictions of the fixed sequence in the SRT task was not significantly different when comparing the goal priming condition ($M = 4.0$, $SD = 1.9$) and the control condition ($M = 3.4$, $SD = 2.0$), $t(156) = 1.949$, $p = .053$, $d = -0.31$.

Additional analyses

3.1. Original key statistical result and interpretation (“claim”)

“As hypothesized, primed participants learned more than participants in the control group, $t(84) = 2.09$, $p_{rep} = .92$, $d = 0.45$ ” (Eitam et al., 2008, p. 265).

“The results of the second experiment, which used the SRT task, replicated this pattern. Goal priming led to improved learning, even though learning was incidental and implicit” (Eitam et al., 2008, pp. 265-266).

3.2. Comparison of Effect Sizes Between Original Study and Replication

Table 3

Comparison of Statistics Between Original Study and Replication

	N	Test statistic	p-value	Test tails	Effect Size Value (95% C.I.)	Effect Size Type	Calculation
Hypothesis: implicit learning as indexed by the difference in response time between the test fixed block and the test random blocks would be greater in the achievement prime condition compared to the control condition.							
Original study	86	$t(84) = 2.09$	.040*	Two	0.45 (.02, .88)**	d	p. 265
Replication study	158	$t(156) = 1.32$	.189	Two	0.21 (-.10, .52)	d	

*The authors reported a p_{rep} value in the original research. This value has been calculated by the replication authors using the t value and df reported in the original research.

**The authors did not report a confidence interval so the 95% confidence interval was calculated by the replication authors using the means, standard deviations, and sample sizes reported in the original research.

3.3. Further results

As noted above, the authors preregistered an additional analysis in an a priori manner. Specifically, SRT learning was also examined using a 2 (condition: goal priming vs. control) x 2 (block type: test fixed vs. test random) ANOVA looking at response times, where condition is a between-subjects factor and block type is a within-subjects factor. There

was not a significant main effect of condition, $F(1, 156) = 0.85, p = .359, \eta^2 = .01$. Not surprisingly, there was a large, significant effect of block type on response times, $F(2, 312) = 20.14, p < .001, \eta^2 = .11$. Examination of this effect of block type revealed that the average response time during the two random blocks was significantly slower than that of the fixed sequence block, $F(1, 156) = 45.35, p < .001, \eta^2 = .23$; in other words, implicit learning had occurred for participants in both conditions. However, the hypothesized block by condition interaction was not significant, $F(2, 312) = 0.88, p = .42, \eta^2 = .01$.

3.4. Dataset and Materials link

SPSS dataset with raw and processed data as well as SPSS syntax for analyses and SPSS output for results can be found at <https://osf.io/br2id/overview>.

3.5. Object/file name

The exact name of the dataset used to arrive at the replication results is SPSS dataset replication of Study 2 Eitam et al 2008 PS.sav.

3.6. Limits to sharing

There are no limits to sharing and all data and materials can be found at <https://osf.io/edcr7>.

3.4. FORRT Replication Database (FReD) Entry

The results of the current replication have already been entered into the FORRT Replication Database.

4. Evaluation

Please carefully describe everything you learned from your replication. This may include acknowledging strengths and limitations of the original or replication study or materials, recommendations for future research on the topic, or how the findings are related to other studies in the area or previous replication studies. Make sure to use neutral and descriptive language.

The current study did not replicate the primary finding from Eitam and colleagues' (2008) experiment 2 that individuals who engaged in a nonconscious goal priming task performed better on an implicit learning task than those participants who did not engage in a goal priming task. The direction of the expected effect was actually reversed, with the control group having a larger mean learning score than that of the goal priming group; however, the difference between the groups was not statistically significant. Gamble et al. (2014) examined the time course of learning differences between a goal priming group and control group and found that there was no difference between the two groups initially, then the goal priming group outperformed the control group, then there was no difference between them again, then the control group outperformed the goal priming group, and then there was no difference between them for the remainder of the experiment. They saw a similar pattern in two experiments that they ran, but the findings were not statistically significant in the first experiment. Also, although the pattern was the same, the time period they saw the differences were not the same for the two experiments (Gamble et al., 2014).

Thus, it is possible that the goal priming condition in the present study would have outperformed the control condition at some time point, but that unlike the original study (Eitam et al., 2008), that time point did not coincide with when the test of learning occurred. There are a few differences between the current replication and the original study that might

have contributed to changing when the goal priming effect materialized; these differences may have contributed to the failure to replicate the primary result of the original study in other ways too.

Probably the most obvious differences between the two studies were cultural ones related to where the study was carried out and who participated in it. Specifically, the original study was conducted in Israel, with participants who were attending school there, and using materials presented in Hebrew, which was likely the native language of most if not all of the participants. Conversely, the replication was carried out in the United States, with participants attending school in the United States, using materials presented in English, which was likely participants' native language. While the replication materials were translated by someone who was fluent in both Hebrew and English, it is still possible that linguistic and cultural differences played a role in the different outcomes of the two studies. Even though efforts were made to assure equivalency between the original Hebrew words and the English words used in the replication, it is possible that the cultural meaning of the achievement words was more profound in the context of the Hebrew language in the Israeli culture relative to the English language in the American culture.

Also, while participants in both studies were enrolled in an institution of higher learning, the average age of participants in the original study ($M = 23.9$ years) was higher than the average age of the replication study participants ($M = 19.65$, $SD = 1.1$). Another demographic difference was a higher proportion of female participants in the replication study (74.8%) as compared to the original study (60.5%). It is possible that cultural, demographic, or the interaction of the two contributed to the failure to replicate the original study's primary finding of NCGP resulting in greater implicit learning. Additional research is needed to better understand the conditions under which nonconscious goal pursuit impacts

implicit learning. Specifically, the extent to which these effects may differ based on age, gender, and/or culture as well as the extent to which these effects are impacted by time.

While the primary result from the original study was not replicated, we did replicate the long-standing finding that the implicit learning of complex patterns can occur without intention or awareness. Consistent with the original study (Eitam et al., 2008), all participants regardless of condition demonstrated significant implicit learning: their mean response times during the fixed test blocks were significantly faster than their mean response times during random blocks. Thus, our results are consistent with a large body of research which shows that humans are able to acquire and apply complex knowledge without intention or awareness.

4.1. Feedback from the original study's authors

As part of the Reproducibility Project: Psychology (Open Science Collaboration, 2015), the authors of the original study were asked if they would like to provide any commentary on our replication attempt and that this commentary would be uploaded as a supplementary file on our OSF website. The authors took us up on this offer and their reply can be found at <https://osf.io/edcr7> (reaction to replication final.docx).

4.2. Limitations and Recommendations for Future Replications

As indicated above, the generalizability of this study's finding is limited by the demographic and cultural make-up of its participants. Future work could examine the extent to which demographic and cultural variables might be moderators of the relation between NCGP and implicit learning. Also, aside from the present study, only one other study that we have been able to find has attempted to replicate Eitam et al.'s (2008) finding and while one of their two experiments found significant effects, they were transient and the effect actually

reversed (also in a transient manner). Therefore, it would also be important to do more research examining the time course of the impacts of NCGP.

4.3. Brief Evaluation

The replication was an informative failure to replicate: the replication study worked as planned, but findings did not confirm the original results.

Contribution Statement

Jason M. Prenoveau: Conceptualization, Data curation, Formal analysis, Investigation, Methodology, Project administration, Resources, Software, Supervision, Writing: original draft, and Writing: review & editing

Matthew W. Kirkhart: Conceptualization, Investigation, Methodology, Project administration, Resources, Supervision, Writing: original draft, and Writing: review & editing

Ethical Statement

All procedures performed in this study, which involved human participants, were in accordance with APA ethical principles and the ethical standards of the 1964 Helsinki Declaration. The study was approved by the Institutional Review Board of Loyola University of Maryland and informed consent was obtained from all study participants.

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Conflicts of interest

The authors declare no conflict of interest associated with the publication of this manuscript.

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Figure 1. Response times of the goal-priming and control groups in the three test blocks.

